For Examiner's Use

9 **(a)** Fig. 9.1 below shows a Rutherford scattering experiment in which α-particles are directed at a gold foil. The detector is shown in two positions in the evacuated chamber.

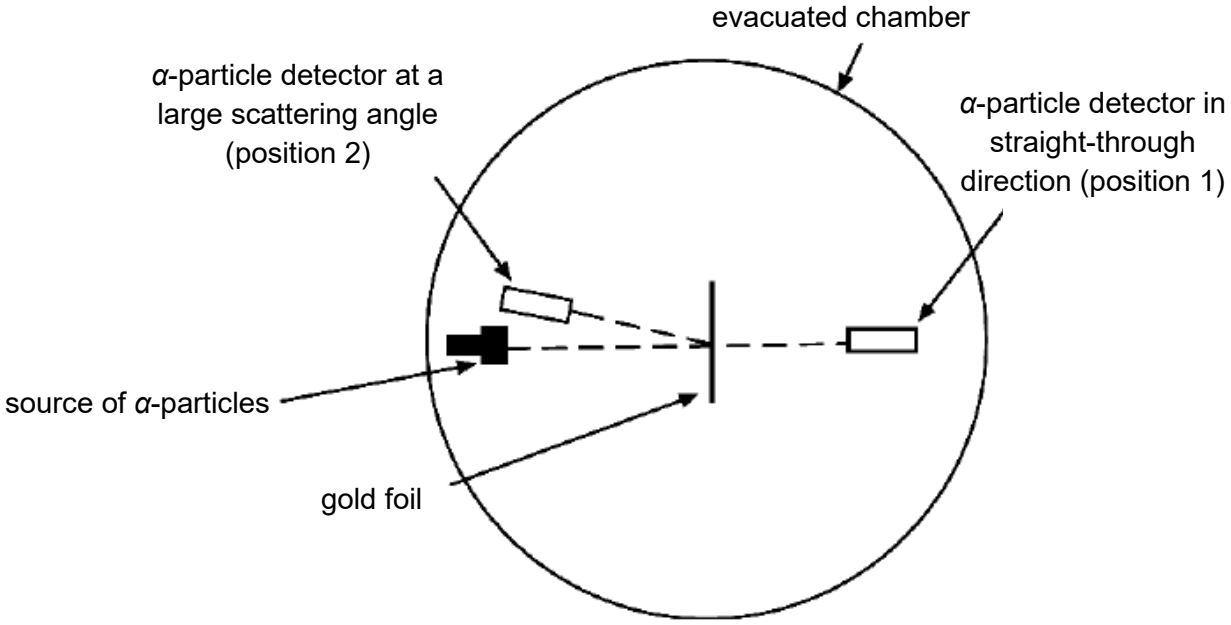

Fig. 9.1

(i) Explain why air needs to be removed from the apparatus.

.....

..... [1]

(ii) Explain why the gold foil should be very thin.

.....

..... [1]

(iii) State what can be deduced, from the following observations, about the structure of the atom and the properties of the gold nucleus:

1. A high count rate is detected by the α-particle detector in position 1.

.....

..... [1]

2. A low count rate is detected by the α-particle detector in position 2.

.....

.....

..... [2]

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(b) Fig. 9.2 illustrates the variation with nucleon number of the binding energy per nucleon of nuclei.

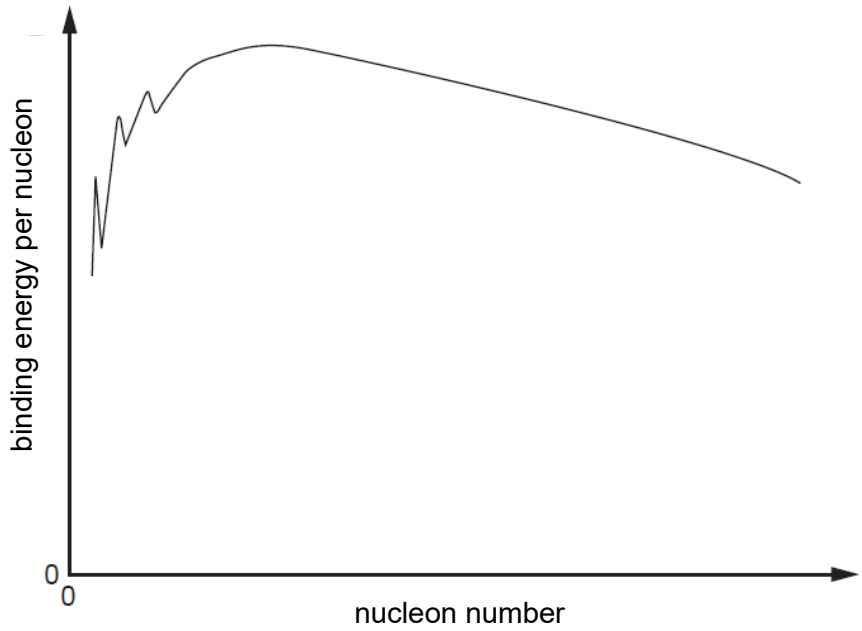

Fig. 9.2

(i) Explain what is meant by the *binding energy* of a nucleus.

.....

.....

..... [2]

(ii) With the aid of Fig. 9.2, explain why more energy per nucleon is released in fusion than in fission.

.....

.....

.....

.....

..... [3]

(iii) On Fig. 9.2, mark with the letter **S** the region of the graph representing nuclei having the greatest stability. [1]

(iv) On Fig. 9.2, complete the graph for elements of low nucleon numbers. [1]

For Examiner's Use

(c) Uranium-235 may undergo fission when bombarded by a slow-moving neutron to produce xenon-142 and strontium-90 as shown below.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{142}_{54}Xe + ^{90}_{39}Sr + \text{neutrons} + ^{0}_{-1}e$$

(i) Determine the number of neutrons produced in this fission reaction.

number = ... [1]

(ii) State the role of these neutrons in the fission reaction.

.....

..... [1]

(iii) Data on binding energy per nucleon for some nuclides is given in Fig. 9.3.

isotope	binding energy per nucleon / MeV
Uranium-235	7.59
Xenon-142	8.23
Strontium-90	8.70

Fig. 9.3

Calculate

1. the energy released in this fission reaction,

energy = ... J [2]

2. the mass equivalent of this energy.

mass = ... kg [1]

For Examiner's Use

(iv) A nuclear power station supplies 100 MW of electrical power to a town for one year at an efficiency of 35%.

Calculate the mass of uranium needed to operate the power station for one year.

mass = …… kg [3]